

***PJM Generation Interconnection Request
System Impact Study Report
For
Queue Position AD1-031
“Kewanee – E.D. Edwards 138 kV”***

September 2021

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner. In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed. The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study. The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D. An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

General

Queue AD1-031 project is a Payton Solar, LLC (“Interconnection Customer” or “IC”) proposal to connect a 70 Energy (26.6 MW Capacity) solar PV facility to be located in Henry County, IL. The IC had proposed a service date for this project of 12/31/2019. This study does not imply a TO commitment to this in-service date.

Point of Interconnection

The AD1-031 will interconnect with the ComEd transmission system via a tap of the Kewanee to E.D Edwards 138 kV line.

ComEd proposes to build a new 138kV substation ‘Kentville Road TSS 962’ adjacent to the Kewanee substation by tapping 138kV line 7423, to interconnect AD1-031.

Cost Summary

Description	Cost Estimate
Total Physical Interconnection Costs	\$ 19,500,000
Allocation towards System Network Upgrade Costs (PJM Identified - Summer Peak)*	\$ 0
Allocation towards System Network Upgrade Costs (PJM Identified - Light Load)*	\$ 0
Allocation towards System Network Upgrade Costs (TO Identified)*	\$ 0
Total Costs	\$ 19,500,000

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

Description	Cost Estimate
Attachment Facilities	\$1,000,000
Direct Connection Network Upgrades	\$ 17,000,000
Non-Direct Connection Network Upgrades	\$1,500,000
Total Physical Interconnection Costs	\$ 19,500,000

Attachment Facilities

The IC AD1-031 generator lead would interconnect to a new 138kV Interconnection Substation. This interconnection would require one 138kV line MOD, a dead-end structure and revenue metering as shown in the one-line diagram.

The cost for the attachment facilities is estimated at \$1M.

Scope of Work	Cost Estimate
Installation of one 138kV line MOD, one dead-end structure and one set of revenue metering (see notes below on cost estimate)	\$1,000,000

Direct Connection Network Upgrades

To accommodate interconnection of AD1-031, a new 138kV Interconnection Substation would need to be built adjacent to Kewanee substation by tapping ‘Kewanee-Edwards (Ameren) 138kV Line 7423.

The scope of work includes the installation of three 138kV circuit breakers in a “breaker-and-a-half” bus configuration and cutting in the Interconnection Substation to the Kewanee-Edwards (Ameren) 138kV Line 7423, as shown in the one-line diagram below.

The IC is responsible for constructing all of the facilities on the IC side of the point of interconnection outside of the substation. It will be the IC’s responsibility to obtain the site for the Interconnection Substation and right-of-way between the Interconnection Substation and the 138kV transmission line.

In the event that the IC exercises the option to build the interconnecting substation, the IC will be required to construct all interconnection facilities that will be turned over to ComEd in accordance with ComEd published standards. The IC will be responsible for the ComEd oversight costs (i.e. costs incurred by the Transmission Owner when engaging in oversight activities to satisfy itself that the IC is complying with the Transmission Owner’s standards and specifications for the construction of facilities).

ComEd would design, engineer and construct the tie in of the Interconnection Substation to the Kewanee-Hennepin (Ameren)-Streator 138kV Line 7423.

The preliminary cost estimate for Direct Connection Network Upgrade is given in the following tables.

For Option to Build Direct Connection cost estimates:

Scope of Work	Cost Estimate
Installation of a new 138kV substation as described above	N/A
Transmission line tie in work (foundations, structures, conductors)	\$ 2,000,000
ComEd oversight and testing	\$ 1,000,000
Total Cost Estimate (see notes below on cost estimate)	\$ 3,000,000

For ComEd building the interconnecting substation cost estimates:

Scope of Work	Cost Estimate
Installation of a new 138kV substation as described above	\$ 15,000,000
Transmission line tie in work (foundations, structures, conductors)	\$ 2,000,000
Total Cost Estimate (see notes below on cost estimate)	\$ 17,000,000

ComEd would take approximately 24-months to construct the substation and transmission line work after the ISA / ICSA are signed.

Non-Direct Connection Network Upgrades

The integration of the new 138kV Interconnection Substation would require relay/communications/SCADA upgrades at TSS 74 Kewanee, TSS 61 Streator, and AMEREN's Hennepin Substation. The ComEd cost is given below:

Scope of Work	Cost Estimate
Relay/communications/SCADA upgrades at TSS 74 Kewanee substation	\$750,000
Relay/communications/SCADA upgrades at E D Edwards Substation (Ameren to provide cost estimate)	\$750,000 ¹
Total Cost Estimate (see notes below on cost estimate)	\$1,500,000

Notes on Cost Estimate:

- 1) These estimates are Order-of-Magnitude estimates of the costs that ComEd would bill to the customer for this interconnection. These estimates are based on a one-line electrical diagram of the project and the information provided by the IC.
- 2) There were no site visits performed for these estimates. There may be costs related to specific site related issues that are not identified in these estimates. The site reviews will be performed during the Facilities Study or during detailed engineering.
- 3) These estimates are not a guarantee of the maximum amount payable by the IC and the actual costs of ComEd's work may differ significantly from these estimates. Per the PJM Tariff, IC will be responsible for paying all actual costs of ComEd's work.
- 4) The IC is responsible for all engineering, procurement, testing and construction of all equipment on the IC's side of the Point of Interconnection (POI).

¹ A preliminary estimate is included in the table. The Interconnection Customer will need to enter into an Affected System study with MISO to evaluate Ameren scope of work.

Network Impacts

The Queue Project AD1-031 was evaluated as a 70.0 MW (Capacity 26.6 MW) injection into a tap of the Kewanee – Edwards 138 kV line in the ComEd area. Project AD1-031 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-031 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Overload Number	Contingency Type	Contingency Name	Affected Area	Facility Description	Bus From	Bus To	Circuit	Power Flow	Loading % Initial	Loading % Final	Rating Type	Rating MVA	MW Contribution	Flowgate Appendix
1	LFFB	COMED_P4_001-38-TR81___	CE - MISO AMIL	HENNEEPIN; T-4HENNEPIN S 138 kV line	271655	348918	1	AC	143.14	146.62	ER	160	6.01	1
2	BUS	COMED_P2-2_001_LA-138B__1	CE - MISO AMIL	HENNEEPIN; T-4HENNEPIN S 138 kV line	271655	348918	1	AC	143.14	146.62	ER	160	6.01	
3	LFFB	COMED_P4_001-38-L0108__	CE - MISO AMIL	HENNEEPIN; T-4HENNEPIN S 138 kV line	271655	348918	1	AC	143.14	146.62	ER	160	6.01	

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

No overdutied breakers.

Affected System Analysis & Mitigation

MISO Impacts:

No MISO system reinforcements were identified to mitigate system violations. See page 6 of this report for Ameren scope for the physical interconnection.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

Overload		Contingency			Bus				Loading %		Rating		
Number	Type	Name	Affected Area	Facility Description	From	To	Circuit	Power Flow	Initial	Final	Type	MVA	MW Contribution
4	N-1	COMED_P1-2_KEWANEE	CE - CE	CRESCENT ; R-OGLESBY ; T 138 kV line	271241	272189	1	AC	109.67	114.95	ER	174	9.46
5	N-1	COMED_P1-2_138-L0112_B-S	CE - MISO AMIL	HENNEEPIN; T-4HENNEPIN S 138 kV line	271655	348918	1	AC	142.81	146.3	ER	160	6.01
6	N-1	COMED_P1-3_TR81_LASCO_B-S	CE - CE	LASCO STA; B-MAZON ; B 138 kV line	271908	271986	1	AC	113.88	115.83	ER	223	4.06
7	N-1	COMED_P2-1_187-L15508__	CE - CE	ROCK FALL; R-NELSON ; R 138 kV line	272367	272095	1	AC	157.09	163.72	ER	223	14.88
8	N-1	COMED_P2-1_187-L15508__	CE - CE	O09 OP1 138-ROCK FALL; R 138 kV line	293510	272367	1	AC	189.99	197.21	ER	214	15.54
9	Non	Non	CE - CE	O09 OP1 138-ROCK FALL; R 138 kV line	293510	272367	1	AC	124.28	129.36	NR	173	8.93
10	N-1	COMED_P2-1_133-CB_23__	CE - CE	O29-NELSON ;RT 138 kV line	293710	272097	1	AC	158.29	164.14	ER	264	15.56
11	Non	Non	CE - CE	O29-NELSON ;RT 138 kV line	293710	272097	1	AC	124.6	128.84	NR	208	9.02

Light Load Analysis - 2021

Not Required.

System Reinforcements

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

To be determined during the Facilities Study.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

1. To alleviate the HENNEPIN; T-4HENNEPIN S 138 kV line overloads:

ComEd:

SSTE rating is 459 MVA. ALDR rating is 573 MVA. No upgrade is required.

MISO:

MISO-end SE rating is 305 MVA. No upgrade is required.

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

Contingencies

Contingency Name	Description
COMED_P1-2_138-L0112__B-S	CONTINGENCY 'COMED_P1-2_138-L0112__B-S' TRIP BRANCH FROM BUS 271844 TO BUS 271908 CKT 1 / KICKA; B 138 LASCO; B 138 END
COMED_P1-2_KEWANEE	CONTINGENCY 'COMED_P1-2_KEWANEE' DISCONNECT BRANCH FROM BUS 271838 TO BUS 348923 CKT 1 END
COMED_P1-3_TR81_LASCO_B-S	CONTINGENCY 'COMED_P1-3_TR81_LASCO_B-S' TRIP BRANCH FROM BUS 270802 TO BUS 270803 CKT 1 / LASCO STA; B 345 LASCO STA; R 345 TRIP BRANCH FROM BUS 270802 TO BUS 271908

	CKT 1 / LASCO STA; B 345 LASCO STA; B 138 END
COMED_P2-1_133-CB_23__	CONTINGENCY 'COMED_P2-1_133-CB_23__' TRIP BRANCH FROM BUS 272367 TO BUS 293510 CKT 1 / R FAL; R 138 O9 138 END
COMED_P2-1_187-L15508__	CONTINGENCY 'COMED_P2-1_187-L15508__' TRIP BRANCH FROM BUS 293710 TO BUS 272097 CKT 1 / O29 ; 138 NELSO;RT 138 END
COMED_P2-2_001_LA-138B__1	CONTINGENCY 'COMED_P2-2_001_LA-138B__1' DISCONNECT BUS 271908 / LASCO STA; B 138 END
COMED_P4_001-38-L0108__	CONTINGENCY 'COMED_P4_001-38-L0108__' TRIP BRANCH FROM BUS 271908 TO BUS 271986 CKT 1 / LASCO; B 138 MAZON; B 138 DISCONNECT BUS 271908 / LASCO; B 138 END
COMED_P4_001-38-TR81__	CONTINGENCY 'COMED_P4_001-38-TR81__' TRIP BRANCH FROM BUS 270802 TO BUS 270803 CKT 1 / LASCO STA; B 345 LASCO STA; R 345 TRIP BRANCH FROM BUS 270802 TO BUS 271908 CKT 1 / LASCO STA; B 345 LASCO STA; B 138

	DISCONNECT BUS 271908 / LASCO STA; B 138 TRIP BRANCH FROM BUS 270802 TO BUS 270803 CKT 1 / LASCO STA; B 345 LASCO STA; R 345 END
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Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the Appendices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the Appendices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators.

It should be noted the project/generator MW contributions presented in the body of the report and appendices sections are full contributions, whereas the loading percentages reported in the body of the report, take into consideration the commercial probability of each project as well as the ramping impact of "Adder" contributions.

Appendix 1

(CE - MISO AMIL) The HENNEEPIN; T-4HENNEPIN S 138 kV line (from bus 271655 to bus 348918 ckt 1) loads from 143.14% to 146.62% (AC power flow) of its emergency rating (160 MVA) for the line fault with failed breaker contingency outage of 'COMED_P4_001-38-TR81____'. This project contributes approximately 6.01 MW to the thermal violation.

CONTINGENCY 'COMED_P4_001-38-TR81____'

TRIP BRANCH FROM BUS 270802 TO BUS 270803 CKT 1 / LASCO STA; B
345 LASCO STA; R 345

TRIP BRANCH FROM BUS 270802 TO BUS 271908 CKT 1 / LASCO STA; B
345 LASCO STA; B 138

DISCONNECT BUS 271908 / LASCO STA; B 138

TRIP BRANCH FROM BUS 270802 TO BUS 270803 CKT 1 / LASCO STA; B
345 LASCO STA; R 345

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
934051	AD1-031 C O1	2.28
934052	AD1-031 E O1	3.72
LTF	BLUEG	0.09
294401	BSHIL;1U E	6.99
294410	BSHIL;2U E	6.99
LTF	CALDERWOOD	< 0.01
LTF	CANNELTON	0.01
LTF	CARR	0.02
LTF	CATAWBA	< 0.01
LTF	CBM-S1	0.02
LTF	CBM-W1	2.16
LTF	CBM-W2	0.75
LTF	CHEOAH	< 0.01
LTF	CLIFTY	0.37
LTF	EDWARDS	0.41
LTF	ELMERSMITH	0.03
LTF	G-007	0.05
LTF	GIBSON	0.03
274871	GR RIDGE ;2U	3.72
274847	GR RIDGE ;BU	2.93
LTF	HAMLET	0.01
990901	L-005 E	9.22
LTF	MEC	1.95
293061	N-015 E	63.97
LTF	NEWTON	0.04
LTF	O-066	0.16
294392	P-010 E	81.24

<i>LTF</i>	<i>RENSSELAER</i>	<i>0.01</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.08</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.02</i>
<i>LTF</i>	<i>WEC</i>	<i>0.13</i>
<i>917451</i>	<i>Z2-081</i>	<i>1.49</i>
<i>925581</i>	<i>AC1-033 C</i>	<i>1.14</i>
<i>925582</i>	<i>AC1-033 E</i>	<i>7.61</i>
<i>926821</i>	<i>AC1-168 C</i>	<i>7.66</i>
<i>926822</i>	<i>AC1-168 E</i>	<i>51.38</i>

Attachment 1: Single Line Diagram

